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The MHV2 Model: A UNIFAC-Based Equation of State Model for Prediction of Gas Solubility and Vapor-Liquid Equilibria at Low and High Pressures

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MHV2 is a group-contribution equation of state based on the modified Huron-Vidal mixing rule. It combines the SRK equation of state with a model for the excess Gibbs energy, the modified UNIFAC model. The MHV2 model can be used to predict vapor-liquid equilibria (VLE) and gas solubility at low and high pressures. Useful predictions for process calculations can be made over a wide temperature and pressure range. This work considers VLE for mixtures with gases. It describes the main features of the MHV2 model and presents tables containing parameters between 9 modified UNIFAC solvent groups and 13 gases (H_2 , O_2 , N_2 , CO , CO_2 , H_2S , CH_4 , C_2H_2 , C_2H_4 , C_2H_6 , C_3H_6 , C_3H_8 , and C_4H_{10}). The gas-gas interaction parameters are all assigned the value 0. It is shown that the MHV2 model compares very well with a density-dependent local composition model (DDLC) and the group-contribution equation of state (GC-EOS). Even for complex systems the multicomponent vapor-liquid equilibria are predicted well.

Introduction

Many systems treated in the chemical industry involve mixed solvents and supercritical components. Calculation of high-pressure vapor-liquid equilibria for such complex systems requires a model that can represent a large number of chemically different systems at a wide range of conditions. This paper presents a model that is applicable to the prediction of binary and multicomponent vapor-liquid equilibrium compositions over a wide range of temperatures and pressures. The model is called the modified Huron-Vidal second-order model (MHV2) and uses an approach developed by Michelsen (1990a, 1990b), in which a cubic equation of state is matched with an excess Gibbs energy model at zero pressure. This approach leads to an explicit Huron-Vidal (Huron and Vidal, 1979) type mixing rule for the a parameter in an equation of state like the Soave Redlich Kwong (SRK) equation (Soave, 1972).

The capability of this model to reproduce low-pressure information stored in excess Gibbs energy models has been illustrated (Michelsen, 1990a, 1990b). Furthermore, the ability of MHV2 together with the modified UNIFAC parameter table (Larsen et al., 1987) for prediction of a high-pressure vapor-liquid equilibrium has been illustrated by Dahl and Michelsen (1990). In the present paper we demonstrate that MHV2 is also able to correlate and predict vapor-liquid equilibria of gas-solvent binary systems and to predict VLE for multicomponent mixtures using the new parameters for gas-solvent interactions together with the modified UNIFAC group-interaction parameter table of Larsen et al. We also discuss precautions to be observed when using the parameter table.

Modified Huron-Vidal Mixing Rule

Huron and Vidal (1979) determined the a parameter of a cubic equation of state from an excess Gibbs energy

model for the liquid state, applied to infinite pressure. The match at infinite pressure renders it impossible to use excess Gibbs energy model parameters based on low-pressure VLE information. The modified Huron-Vidal mixing rule is based on a similar approach, but the equation of state is matched with the excess Gibbs energy model at zero pressure, where the existing parameters based on low- or normal-pressure VLE information may be applied. In this work we have used the SRK equation of state and the modified UNIFAC model.

The SRK equation of state may be written as follows:

$$P = \frac{RT}{v-b} - \frac{a}{v(v+b)} \quad (1)$$

where the mixture b parameter is derived from the conventional linear mixing rule

$$b = \sum_{i=1}^c z_i b_{ii} \quad (2)$$

using for the corresponding pure component parameter

$$b_{ii} = 0.08664 RT_{ci} / P_{ci} \quad (3)$$

The pure component a parameter is given by

$$a_{ii} = 0.4286 (R^2 T_{ci}^2 / P_{ci}) [f(T_{ri})]^2 \quad (4)$$

with (Mathias and Copeman, 1983)

$$f(T_{ri}) = 1 + C_1(1 - (T_{ri})^{1/2}) + C_2(1 - (T_{ri})^{1/2})^2 + \frac{C_3(1 - (T_{ri})^{1/2})^3}{T_{ri}} \quad T_{ri} < 1$$

$$= 1 + C_1(1 - (T_{ri})^{1/2}) \quad T_{ri} > 1 \quad (5)$$

The pure-component vapor pressures used to estimate C_1 , C_2 , and C_3 were calculated by using the AIChE DIPPR compilation constants (Daubert and Danner, 1986). Values of the C parameters used in this work are given in Table

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Table I. Pure-Component Temperature Parameters (Eq 5)

component	C_1	C_2	C_3
hydrogen	0.1334	0.0000	0.0000
oxygen	0.5456	-0.2451	0.3385
nitrogen	0.5427	-0.0524	-0.3381
carbon monoxide	0.5836	0.0000	0.0000
carbon dioxide	0.8653	-0.4386	1.3447
hydrogen sulfide	0.5507	0.4534	-0.5841
methane	0.5472	-0.3992	0.5751
acetylene	0.9071	-1.5975	4.5697
ethene	0.8479	-0.3421	0.6603
ethane	0.6853	-0.4284	0.7382
propene	0.7517	-0.6277	1.7387
propane	0.7726	-0.5090	1.0306
butane	0.8487	-0.5520	1.0774
pentane	0.9372	-0.7656	1.8478
hexane	1.0827	-1.2797	2.6177
heptane	1.0874	-0.7465	1.5765
octane	1.0736	0.0656	-0.2720
nonane	1.1365	0.0786	-0.3464
decane	1.1406	0.5867	-1.3610
1-butene	0.8389	-0.9122	2.8447
2-methyl-1-pentene	0.8373	0.0117	0.3814
1-hexene	0.9408	-0.3064	0.6032
1-heptene	1.0478	-0.5869	1.1697
1-octene	1.0844	-0.3636	0.7564
benzene	0.8457	-0.3856	0.8843
toluene	0.9363	-0.4585	0.8674
ethylbenzene	0.9918	-0.4355	0.8143
p-xylene	1.1190	-1.2050	2.1300
m-xylene	1.0285	-0.4432	0.6927
methanol	1.4450	-0.8150	0.2486
ethanol	1.4252	0.1898	-1.3014
1-propanol	1.2645	1.2138	-2.4077
2-propanol	1.3542	0.9546	-2.1524
2-methyl-1-propanol	1.2048	0.8148	0.3299
2-methyl-2-propanol	1.2064	1.1632	-0.4743
1-butanol	1.1345	1.7856	-2.8325
2-butanol	1.0175	2.4045	-3.4954
1-pentanol	1.3138	0.0962	0.9660
1-hexanol	1.2187	0.8378	-0.6983
water	1.0873	-0.6377	0.6345
acetone	0.9998	-0.4137	0.5345
butanone	1.0429	-0.5621	0.7275
methyl acetate	1.0232	-0.4357	0.8420
ethyl butyrate	1.1490	-0.3372	0.0873
ethyl acetate	1.1197	-0.6744	1.3238
butyl acetate	0.9940	0.9555	-2.3004
methyl ether	0.8837	-1.0325	2.8278
ethyl ether	1.0500	-1.3238	3.0278
propyl ether	1.1780	-1.2082	2.2417
butyl ether	1.2158	-0.5675	0.7140
triethylene glycol	2.4920	-1.4627	-2.6264

I. For hydrogen and carbon monoxide, C_1 has been calculated from the acentric factor ω (Soave, 1972):

$$C_1 = 0.48 + 1.574\omega - 0.176\omega^2 \quad (6)$$

A description of the model equations including a procedure for the calculation of fugacity coefficients is given by Dahl and Michelsen (1990). The MHV2 mixing rule is

$$q_1(\alpha_{\text{mix}} - \sum_{i=1}^C z_i \alpha_{ii}) + q_2(\alpha_{\text{mix}}^2 - \sum_{i=1}^C z_i \alpha_{ii}^2) = \frac{g^E}{RT} + \sum_{i=1}^C z_i \ln \left(\frac{b}{b_{ii}} \right) \quad (7)$$

where $\alpha_{\text{mix}} = a_{\text{mix}}/bRT$ and $\alpha_{ii} = a_{ii}/b_{ii}RT$. Dahl and Michelsen determined the values of q_1 and q_2 : $q_1 = -0.478$ and $q_2 = -0.0047$. Since all the parameters in eq 7 are known and since g^E can be calculated from the g^E model (in this case modified UNIFAC), we can solve eq 7 for α_{mix} yielding the mixture parameter for the equation of state.

Table II. Molecular R_v and Q values and the Main-Group (MG) Definitions for the Gases

gas	MG	R_v	Q
H ₂	22	0.8320	1.1410
O ₂	23	1.7640	1.9100
N ₂	24	1.8680	1.9700
CO	25	2.0940	2.1200
CO ₂	26	2.5920	2.5220
H ₂ S	27	2.3330	2.3260
CH ₄	28	2.2440	2.3120
C ₂ H ₂	29	3.0454	2.7840
C ₂ H ₄	30	3.1482	2.9700
C ₂ H ₆	31	3.6044	3.3920
C ₃ H ₆	32	4.4930	4.0480
C ₃ H ₈	33	4.9532	4.4720
C ₄ H ₁₀	34	6.3020	5.5520

Equation 7 is used under all conditions for both correlations and predictions.

The fugacity coefficient is given by

$$\ln \varphi_i = \ln \left(\frac{RT}{P(v-b)} \right) + \left(\frac{1}{v-b} - \frac{\alpha}{v+b} \right) b_{ii} - \ln \left(\frac{v+b}{v} \right) \left(\frac{\partial(n\alpha)}{\partial n_i} \right)_{T,n_j} \quad (8)$$

where the composition derivative of $n\alpha$ can be calculated from the mixing rule, eq 7.

For the modified UNIFAC model, interaction parameters between 45 subgroups and 21 main solvent groups have been determined by Larsen et al. (1987). In the present work we have added 13 new groups for systems involving gases. The group definitions for these new groups together with the structural parameters R_v and Q (Sander et al., 1983) are listed in Table II.

Models Used for Comparison. The MHV2 model was compared with the DDLC model (Mollerup and Clark, 1989), which is based on the density-dependent local composition theory. This model is capable of describing complex polar and nonpolar systems at varying density levels. Because the a parameter for MHV2 is density-independent, it is of interest to compare the behavior of MHV2 with that of the DDLC model. We have also used the group-contribution equation of state (GC-EOS, Skjold-Jørgensen, 1988) for comparison as this model has extensive, readily available parameter tables. It should be noted that MHV2 is computationally faster than both the DDLC and the GC-EOS models.

Parameter Tables and Estimation. The modified UNIFAC parameters (a_{ij}) determined in this work for the interactions between the modified UNIFAC main groups 1-7, 9-10 (Larsen et al., 1987), and the 13 gases (main group 22-34) are given in Table III. In the expression used to describe the temperature dependence of the interaction parameters in modified UNIFAC we use only two terms, i.e.

$$a_{ij} = a_{ij,1} + a_{ij,2}(T - T_0) \quad (9)$$

where T_0 is a reference temperature, $T_0 = 298.15$ K, and i and j represent the main groups. Further, all gas-gas interaction parameters have been assigned a value of zero.

In Table IIIa we have given supplementary values of the interaction parameters between the CH₂ group of an alcohol and the gases H₂ and N₂. It was not possible to describe the VLE behavior for N₂ and H₂ by using the same values for the interaction parameters ($a_{\text{CH}_2, \text{gas}}$) of the CH₂ groups present in alkanes and in alcohols. The same problem appeared with the GC-EOS model for which special "water-soluble" CH₂ groups in alcohols and ketones

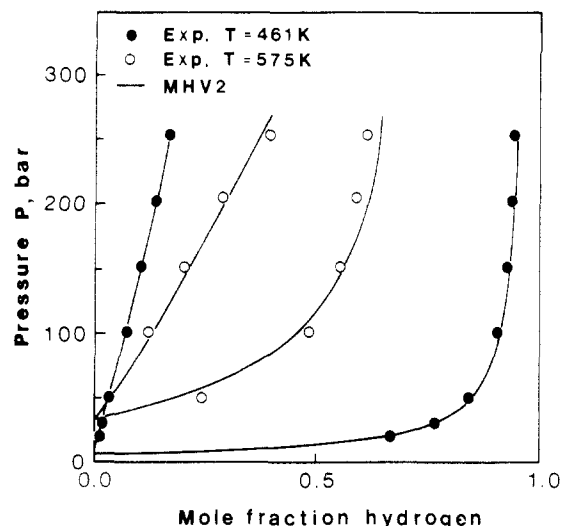


Figure 1. P - XY diagram for hydrogen-toluene at 461 and 575 K. Experimental data: Simnick et al. (1978).

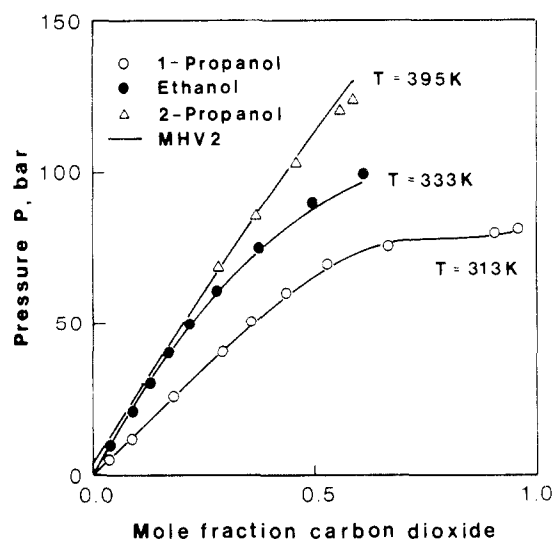


Figure 2. P - X diagram for CO_2 -ethanol at 333 K, CO_2 -1-propanol at 313 K, and CO_2 -2-propanol at 395 K. Experimental data: Radosz (1986); Suzuki et al. (1990).

were introduced. The CH_2 group present in an alcohol is here denoted CH_2' . The $a_{\text{CH}_2', \text{gas}}$ values in Table III should be used when the CH_2 group is in an alcohol molecule. The details of parameter estimation are given in the Appendix.

Temperature Dependence of the Model. We have tested the temperature dependence of the MHV2 model using eq 9 for various binary systems. In Table IV we show VLE results (bubble- P calculations) for the systems H_2 - H_2O , CH_4 - H_2O , C_2H_6 - C_6H_6 , and CO - CH_3OH . For each system, one set of parameters was determined from all the experimental data points available (second line in each system block, Table IV), and another set of parameters was estimated by using data below 423 K (low-temperature parameters). The first line in each system block of Table IV shows the deviations in pressure and vapor-phase composition when the low-temperature parameters are used to calculate VLE including all the experimental data points, and the second line shows the calculated deviations with the parameters estimated by using all the experimental data points. For the hydrogen-water system, the low-temperature parameters were estimated in the interval 311–366 K and then used to predict VLE up to 589 K with no visible degradation of the results. As can be seen from Table IV, the high-temperature predictions are acceptable,

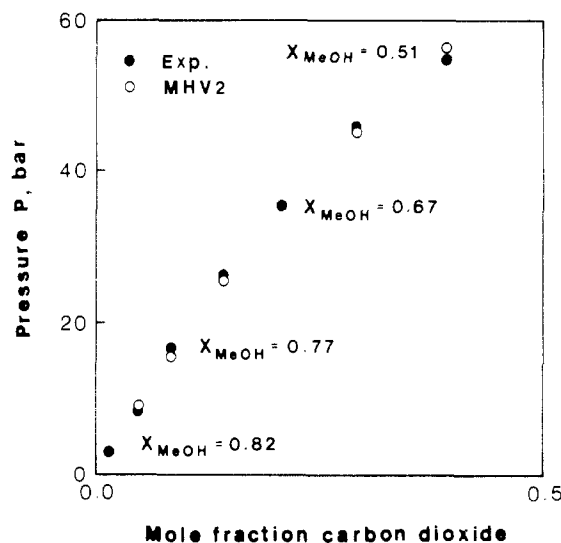


Figure 3. P - XY diagram for CO_2 -methanol-water at 298 K. Two of the experimental data points are identical with the calculated values (X_{MeOH} is the mole fraction of methanol in the liquid phase). Experimental data: Chang and Rousseau (1985).

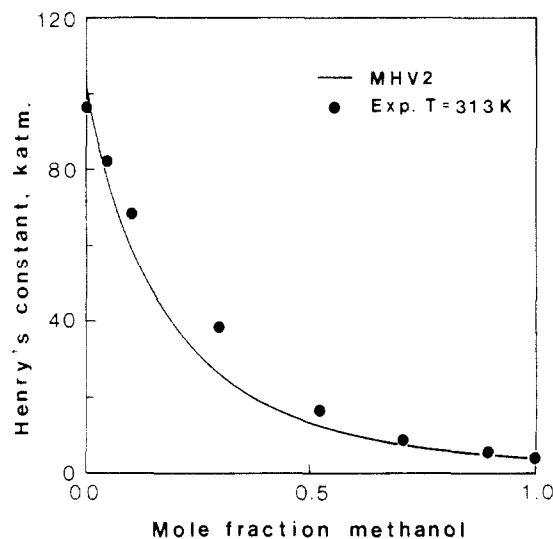


Figure 4. Experimental and predicted Henry's constants for N_2 in methanol-water solutions at 313 K. Experimental data: Tokunaga (1975).

except for the methane-water system.

Binary Systems. A typical example of the VLE behavior of a binary mixture with a gas at high temperatures and pressures is the hydrogen-toluene system, where Figure 1 shows isotherms at 461 and 575 K. Figure 2 shows P - X diagrams for carbon dioxide with three different alcohols. VLE results are presented for three binary systems with water in Table V. It shows the results from flash calculations with DDLC and MHV2 given as the total average deviation of the logarithm of the equilibrium ratios ($\ln(y_i/x_i)$). Only VLE data were used in the estimation of the parameters, and the calculated results for liquid-liquid equilibria are satisfactory compared with the DDLC model for which also LLE data were used in the parameter estimation.

Multicomponent Mixtures. To test the ability of the model to predict VLE for multicomponent mixtures with gases, we have investigated 12 multicomponent systems. Figure 3 shows experimental and calculated pressures in the carbon dioxide-methanol-water system. The MHV2 model is able to cover the concentration range (the maximum content of water is 0.17 mol %) of this system well.

Table III

(a) Modified UNIFAC Interaction Parameters (a_{ij} ; First Row Gives $a_{ij,1}$ and Second Row $a_{ij,2}$ (Eq 9))

i	j												
	22	23	24	25	26	27	28	29	30	31	32	33	34
1	509.0	84.38	279.8	58.78	123.9	188.3	-109.4	24.36	-133.0	-86.55	46.08	35.48	-22.08
CH ₂	1.440	-2.290	1.119	0.000	-0.4065	0.0251	-0.8640	-1.050	-0.8723	-0.7601	-0.1130	-0.0620	-0.5395
2	527.6	na	130.9	17.54	-28.36	na	na	na	-115.9	-173.9	-30.76	-140.1	171.1
C=C	0.3671	na	0.000	0.000	1.169	na	na	na	0.0569	9.200	-0.0052	0.7277	0.5866
3	188.3	147.4	94.66	141.0	134.1	326.9	-3.522	7.493	214.9	1.951	-65.29	2.323	2.323
ACH	3.334	-1.950	1.767	-1.770	1.738	0.805	-0.4008	-0.8400	0.00	-0.2326	-1.521	-0.6311	-0.6311
4	432.1 ^a	675.6	257.2 ^a	446.4	261.7	425.8	503.1	425.8	802.0	514.5	642.5	469.3	524.7
OH	-0.1257	0.0137	0.0089	-2.385	5.381	6.850	0.000	2.309	-5.435	-3.574	0.000	-4.203	-1.664
5	198.9	132.1	57.62	-19.85	-126.6	-132.8	-55.30	30.00	-70.84	-37.64	-93.00	-67.57	-53.35
CH ₃ OH	-2.138	-1.672	-1.615	-0.9319	-0.2024	0.000	-1.104	-0.5165	-0.8463	-0.8812	0.000	-0.7372	-0.8700
6	949.9	688.7	596.3	494.0	226.6	253.9	499.2	na	346.5	405.0	311.5	361.8	331.8
H ₂ O	-0.3100	-0.9018	0.2784	0.1390	-0.2410	-0.3050	-0.2550	na	-0.3326	0.0930	-0.0472	0.1271	-0.2550
7	273.7	124.3	181.4	na	81.51	na	769.5	na	216.2	-53.51	-70.26	-9.088	na
CH ₂ CO	-6.293	-2.543	-2.390	na	0.000	na	0.000	na	0.000	0.000	0.000	-1.490	na
9	319.0	na	217.9	159.0	-80.55	-0.1893	na	na	na	-80.97	na	na	na
CCOO	-5.3620	na	-2.270	-1.039	-0.5914	0.1456	na	na	na	0.000	na	na	na
10	341.3	67.30	262.8	57.04	117.7	152.4	177.5	na	na	308.5	464.3	na	na
CH ₂ O	-7.130	-0.9230	-1.708	-3.694	5.759	-2.750	-1.502	na	na	0.000	0.000	na	na

(b) Modified UNIFAC Interaction Parameters (a_{ij})

	j									
i	1	2	3	4	5	6	7	9	10	
22	-228.1	-227.0	-54.49	1746.0	1540.90	1586.0	273.7	319.0	341.3	
H ₂	-1.304	-1.305	-1.545	5.153	-5.567	3.924	-6.293	-5.362	-7.130	
23	-5.922	na	147.4	1453.0	132.1	688.7	124.3	na	67.30	
O ₂	-0.0372	na	-1.950	4.560	-1.672	-0.9018	-2.5430	na	-0.9230	
24	-152.5	-25.36	121.2	192.9	1405.0	1723.0	181.4	217.9	262.8	
N ₂	-1.241	0.000	-2.259	-3.616	-1.881	-4.652	-2.390	-2.270	-1.708	
25	4.278	17.54	141.0	1170.0	1368.0	1455.0	na	159.0	57.04	
CO	0.000	0.000	-1.770	-9.344	-2.928	-2.906	na	-1.039	-3.694	
26	-55.69	-47.16	-16.56	583.3	727.9	1067.0	-0.0172	153.0	82.87	
CO ₂	-0.4904	0.6526	-1.928	-4.940	-1.331	-0.4180	0.000	0.4509	-2.877	
27	-93.27	na	-196.2	594.7	800.3	753.0	na	-0.1893	152.4	
H ₂ S	-0.6816	na	-0.7045	3.480	0.000	-0.8809	na	0.1456	-2.750	
28	118.0	na	148.7	979.7	1532.0	1608.0	-11.90	na	198.8	
CH ₄	0.3910	na	-0.2084	0.000	-4.137	-2.059	0.000	na	0.5265	
29	24.36	na	7.493	425.8	30.00	na	na	na	na	
C ₂ H ₂	-1.050	na	-0.8400	2.309	-0.5165	na	na	na	na	
30	149.1	152.7	-144.5	802.0	672.6	1354.0	76.87	na	na	
C ₂ H ₄	0.5241	0.2693	0.000	-5.453	-2.160	-1.542	0.000	na	na	
31	99.22	185.0	70.69	663.1	760.9	1529.0	579.2	501.6	138.6	
C ₂ H ₆	0.4350	-5.864	-0.1656	0.0349	1.374	-3.081	0.000	0.000	0.000	
32	-40.85	46.11	433.1	642.5	1250.0	1560.0	469.0	na	89.73	
C ₃ H ₆	-0.2031	-0.0060	0.7479	0.000	0.000	-2.863	0.000	na	0.000	
33	-27.08	234.90	84.60	561.1	1138.0	1644.0	115.1	na	na	
C ₃ H ₈	-0.1935	-1.321	0.3180	-0.9400	1.180	-3.550	3.014	na	na	
34	31.77	-88.39	84.60	524.7	689.8	1606.0	na	na	na	
C ₄ H ₁₀	0.4364	-0.4392	0.3180	-1.664	4.078	-2.221	na	na	na	

^a For the solubility of H₂ and N₂ in alcohols, use the following parameters for the CH₂ groups in the alcohol ($a_{ij,1}$, $a_{ij,2}$): $a_{CH_2, H_2} = 986.0$, -2.133 ; $a_{H_2, CH_2} = -129.2$, -2.418 ; $a_{CH_2, N_2} = 355.2$, 0.000 ; $a_{N_2, CH_2} = 139.4$, 0.000 .

Table IV. VLE Results for MHV2^a

system	params est from			av dev, all data points included		
	T , K	P , bar	no. of dtps	$\Delta P/P \times 100$	$\Delta Y \times 100$	ref
hydrogen-water	311-366	3-138	6	4.1	0.6	Gillespie and Wilson (1980)
	311-589	3-138	13	2.8	0.5	
methane-water	323-423	14-138	9	11.1	1.6	Gillespie and Wilson (1982)
	323-589	14-169	16	4.3	0.7	
ethane-benzene	313-373	20-50	19	4.6	1.1	Knapp et al. (1982)
	313-533	8-83	70	2.6	0.8	
carbon monoxide-methanol	298-373	44-379	17	4.1	1.6	Brunner et al. (1987)
	298-473	44-457	33	3.2	1.0	

^a Predictions obtained with parameters estimated from low temperature information versus results obtained with parameters estimated on the basis of all the experimental data points.

Figures 4 and 5 shows Henry's constant-composition plots for N₂-methanol-water and CO₂-1-propanol-water solutions. The calculated Henry's constants are in reasonable agreement with experiment.

Table VI shows bubble-point calculations for six ternary systems with the MHV2 and GC-EOS models. In all cases the deviations for MHV2 are below 12% in pressure and below 0.04 in composition. For the GC-EOS we usually

Table V. Flash Calculations for Three Systems with Gases in Water, VLE, and LLE and the Mean Deviation $\Delta \ln K^a$

system	type	T, K	P, bar	mean dev	
				DDLC	MHV2
carbon dioxide-water	VLE	302-477	7-202	0.074	0.053
hydrogen	VLE	311-589	8-207	0.077	0.064
sulfide-water	LLE*	311	41-207	0.123	0.152
methane-water	VLE	323-589	14-169	0.110	0.039

LLE = liquid-liquid equilibria predicted with MHV2. Experimental data: Gillespie and Wilson (1982).

$$\text{average } \Delta \ln K = \frac{\sum_{i=1}^{\text{dpts}} \sum_{j=1}^C \left| \frac{\ln K_j(\text{cal}) - \ln K_j(\text{exp})}{C \cdot \text{dpts}} \right|}{K_j = \frac{y_j}{x_j}}$$

C: number of components (C = 2).

obtain somewhat higher deviations, the calculated pressures being too high when more than one gas is present. The reason is probably the use of gas-gas interaction parameters for the GC-EOS model, which at lower temperatures can influence the vapor-phase fugacities strongly. Gas-gas interaction parameters can also be estimated for the MHV2 model (from binary VLE data). When such parameters are used for a system such as nitrogen-carbon dioxide-methanol, the same tendency is observed as with the GC-EOS model.

In Table VII results from flash and dew-point calculations for four multicomponent systems are shown. The deviations have been calculated for the equilibrium ratios (K values) for each component. For system A ($\text{H}_2\text{-CO-H}_2\text{O}$) we have performed dew-point calculations ($\Delta P/P$ for MHV2 is 4.3% and for the DDLC model is 3.6%), and the distribution of all the components is determined accurately with both models. For system B ($\text{CH}_4\text{-CO}_2\text{-H}_2\text{S-H}_2\text{O}$) the K values for methane and hydrogen sulfide were represented better with MHV2 than with the DDLC model, and the deviation is quite low for this system with both models.

Table VI. VLE Results for Ternary Systems (Predictions Obtained from Bubble-Point Calculations)

system	T, K	P, bar	no. of dpts	$\Delta P/P \times 100$	$\Delta y \times 100$	ref
carbon dioxide (1)-1-hexene (2)- <i>n</i> -hexane (3)	303-323	19-86	47	MHV2 7.2	0.3	Wagner and Wichterle (1987)
methanol (1)-carbon dioxide (2)-water(3)	298	3-54	13	GC-EOS 12.4 MHV2 3.1		Chang and Rousseau (1985)
nitrogen (1)-ethylene (2)-methanol (3)	260	50	6	GC-EOS 18.5 MHV2 11.4	3.5 2.9	Zeck and Knapp (1986)
nitrogen (1)-ethane (2)-methanol (3)	260	15-75	20	GC-EOS 14.1 MHV2 8.9	2.5 2.2	Zeck and Knapp (1986)
nitrogen (1)-carbon dioxide (2)-methanol (3)	273	50-125	28	GC-EOS 30.4 MHV2 9.9	2.7 1.4	Weber et al. (1984)
propane (1)-methanol (2)-carbon dioxide (3)	313-343	5-32	54	GC-EOS 7.9 MHV2 8.8	3.4 3.3	Galivel-Solastiouk (1986)

Table VII. Deviations of the Equilibrium Ratios from Flash and Dew-Point Calculations for Four Multicomponent Systems^a

syst	P, bar	T, K	no. of dpts	model	av dev $\ln K$	ref
A	3-138	311-589	15	DDLC	0.08	Gillespie and Wilson (1980)
				MHV2	0.07	
B	48-182	311-450	16	DDLC	0.17	Robinson et al. (1982)
				MHV2	0.13	
C	95	308	1	GC-EOS	0.18	Pan and Ma (1985)
				MHV2	0.12	
D	77-174	313-333	13	GC-EOS	0.31	Suzuki et al. (1990)
				MHV2	0.24	

^aSystem specifications: (A) $\text{H}_2\text{-CO-H}_2\text{O}$, dew-point calculations ($\Delta P/P$ less than 4.3% for both models). (B) $\text{CH}_4\text{-CO}_2\text{-H}_2\text{S-H}_2\text{O}$, flash calculations. (C) $\text{CH}_3\text{OH-H}_2\text{O-CO}_2\text{-CO-H}_2\text{-CH}_4\text{-N}_2$, flash. (D) $\text{H}_2\text{-CO-CO}_2\text{-CH}_4\text{-C}_2\text{H}_6\text{-C}_3\text{H}_8\text{-H}_2\text{O-CH}_3\text{OH-CH}_3\text{CH}_2\text{OH-*n*-CH}_3(\text{CH}_2)_2\text{OH}$, flash. Average deviation:

$$\text{average } \Delta \ln K = \frac{\sum_{i=1}^{\text{dpts}} \sum_{j=1}^C \left| \frac{\ln K_j(\text{cal}) - \ln K_j(\text{exp})}{C \cdot \text{dpts}} \right|}{K_j = \frac{y_j}{x_j}}$$

C: number of components.

For system C the, "methanol synthesis loop", MHV2, is slightly better than GC-EOS. The last system, system D, contains water, three alcohols, and six gases. We obtained larger deviations for this system than for the other systems investigated, in particular for hydrogen and the solvents (water and alcohols). In Figure 6 we have shown the K values for five gases as a function of the total pressure at 333 K for system D. The K values for hydrogen and propane predicted with MHV2 are too low, but the K values for all the other gaseous components are described well.

The Group Contribution Concept

A number of gas-solvent group interaction parameters are determined from only one or two different data sets. In such cases, it is uncertain how good the predictions will be for the other mixtures of the same type. As an example the H_2S -alcohol interaction parameters are based on only one system (H_2S -ethanol), and is uncertain how well H_2S -1-butanol is represented. To indicate the quality of the predictions and to illustrate the value of the group-contribution concept, we have investigated various binary mixtures containing gases for which abundant high-pressure VLE data were available. When two different mixtures contain the same combination of groups (e.g., H_2S -ethanol and H_2S -1-butanol), we define them as belonging to the same class. Table VIII gives the mixture specifications and the parameters used to represent the different classes. Table IX shows the VLE predictions obtained when the parameters from Table VIII are used to represent other systems. The first column in Table IX shows the mixtures of Table VIII (denoted A-E) from which the parameters were obtained. The parameters used for the interaction coefficients between the CH_2 groups and the gases are the values from Table III. The deviations given in parentheses in Table IX are those obtained with all

Table VIII. Systems and Parameters Used in the VLE Predictions, Table IX (First Row Gives $a_{ij,1}$ and Second Row $a_{ij,2}$, Eq 9)^a

params used in Table IX obtained from	T, K	P, bar	no. of dtps	a_{ij}	a_{ji}	j	i	ref
B: CO ₂ -1-butene	273	3-26	9	-158.7 0.000	269.4 0.000	CO ₂	C=C	Knapp et al. (1981)
C: CO ₂ -toluene	311-394	3-138	20	57.08 1.896	44.23 -2.20	CO ₂	ACH	Knapp et al. (1981)
D: methane-ethanol	313-333	18-105	10	869.6 0.000	659.5 0.000	CH ₄	OH	Suzuki et al. (1990)
E: CO ₂ -diethyl ether	298-313	7-72	17	6.81 0.000	209.6 0.000	CO ₂	CH ₂ O	Knapp et al. (1981)

^a A: values for the gas-CH₂ interaction parameters from Table III.

Table IX. VLE Predictions for Binary Systems Using Parameters from Table VIII (The Deviations Given in Parentheses Are Those Obtained with Parameters from Table III)

params from	syst	T, K	P, bar	no. of dtps	$\Delta P/P \times 100$	$\Delta y \times 100$	ref ^a
A	CO ₂ -cyclohexane	473-493	20-75	10	13.1	2.2	1
	CO ₂ -methylcyclohexane	311-477	3-136	23	6.3	1.4	2
	ethane-cyclohexane	283-483	7-83	55	13.3	3.1	1
	methane-cyclohexane	294-378	13-207	19	23.4	1.0	1
B	CO ₂ -2-methyl-1-pentene	303-343	10-61	23	10.6 (5.9)	0.8 (0.8)	3
	CO ₂ -1-hexene	303-343	10-61	25	5.1 (4.8)	0.6 (0.6)	3
	CO ₂ -1-heptene	303-343	10-61	26	8.5 (7.0)	0.3 (0.3)	3
C	CO ₂ -benzene	313-393	1-133	32	3.7 (3.9)		4
	CO ₂ -ethylbenzene	313-366	24-115	14	6.2 (5.0)	0.2 (0.2)	5
	CO ₂ -p-xylene	313-366	11-129	16	8.3 (7.3)	0.4 (0.3)	5
D	methane-1-propanol	313-333	14-101	10	6.2 (4.1)	0.1 (0.1)	6
	CO ₂ -dimethyl ether	273-387	4-73	76	7.3 (5.7)	2.2 (1.9)	7
E	CO ₂ -dibutyl ether	312-366	11-114	20	10.5 (8.9)	0.4 (0.3)	6

^a 1, Knapp et al. (1981); 2, Ng and Robinson (1979); 3, Vera et al. (1984); 4, Gupta et al. (1982); 5, Mohammed and Holder (1987); 6, Suzuki et al. (1990); 7, Tsang and Street (1987).

Table X. Specification of Experimental Information Used in the Estimation of Parameters^a

Mg	H ₂	O ₂	N ₂	CO	CO ₂	H ₂ S	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	C ₃ H ₆	C ₃ H ₈	C ₄ H ₁₀
1. CH ₂	A 2 ^b	C 4 ^b	A 4 ^b	A	A 4 ^b	A 3 ^b	A 5 ^b	C	A 2 ^b	A 5 ^b	A 5 ^b	A 5 ^b	A 3 ^b
2. C=C	A 2 ^b	na ^c	A ^d	A ^e	A 5 ^b	na	na	na	A ^f	A ^g	A ^h	A ⁱ	A ^j
3. ACH	A	C	A	C	A 5 ^b	A 3 ^b	A 3 ^b	D	A	A	A	A	A
4. OH	B 7 ^b	C 2 ^b	C 7 ^b	B 7 ^b	A 3 ^b	B	A 2 ^b	C	C 5 ^b	A 2 ^b	C 4 ^b	A	B 2 ^b
5. CH ₃ OH	A	C	A	A	A	B	A	C	B	A	A	A	B
6. H ₂ O	A	B	A	A	A	A	A	na	A	A	A	A	A
7. CH ₂ CO	A	C	C	na	A	na	A	na	A	A	A	A	na
9. CCOO	C 2 ^b	na	C 2 ^b	C	A 4 ^b	C	na	na	na	A	na	na	na
10. CH ₂ O	C 2 ^b	C 2 ^b	C 2 ^b	C	A 3 ^b	C	A 2 ^b	na	na	A	A	na	na

^a A: TPXY data (case 1). B: TPX data (case 2). C: TX data (case 3). ^b We have listed the number of different mixtures for which experimental data were used in the estimation of parameters, if more than one. ^c na: not available. ^d Parameters obtained from the N₂-C₃H₈ system. ^e Parameters obtained from the CO-C₃H₈ system. ^f Parameters obtained from the C₂H₄-C₃H₈ system. ^g Parameters obtained from the C₂H₆-C₃H₈ system. ^h Parameters obtained from the C₃H₆-1-butene system. ⁱ Parameters obtained from the C₃H₈-C₃H₆ system. ^j Parameters obtained from the C₄H₁₀-1-butene system.

interaction parameters from Table III. The predictions for the first four mixtures, i.e., systems containing cyclic compounds (cyclohexane and methylcyclohexane), were carried out by using interaction parameters between the gases and the CH₂ groups estimated from alkane data. Considering that alkanes and cyclic compounds have different properties, the predictions are in most cases satisfactory. For mixtures containing alkenes we have estimated parameters ($a_{\text{CO}_2, \text{C}=\text{C}}$) from the CO₂-1-butene system (with 1-butene treated as a solvent molecule) at 273 K (Table VIII). These parameters are then used to predict VLE for the mixtures containing CO₂ with 1-methyl-2-pentene, 1-hexene, and 1-heptene. Although only a single isotherm was used in the estimation of parameters for this class of systems, the predictions obtained are satisfactory.

In all the examples given in Table IX the group-contribution concept yields results of good quality. However, the UNIFAC group contribution concept is not able to account for proximity effects, i.e., the influence of a group

on neighbor groups in the same molecule. These effects can influence the properties of a functional group significantly, especially when conjugation effects or intramolecular hydrogen bonds are present.

Conclusions

It has been illustrated that the MHV2 model is able to treat binary and multicomponent systems with gases at both low and high pressures. Many of the systems investigated in this work contain water and polar and non-polar gases, and still the model compares well with the DDLC model. For the multicomponent systems investigated in this work the overall performance of MHV2 is far better than that of GC-EOS. In addition we have shown that with MHV2, phase equilibria can be described without the use of gas-gas interaction coefficients and with a limited number of parameters to represent a very broad temperature interval and a large variety of molecules.

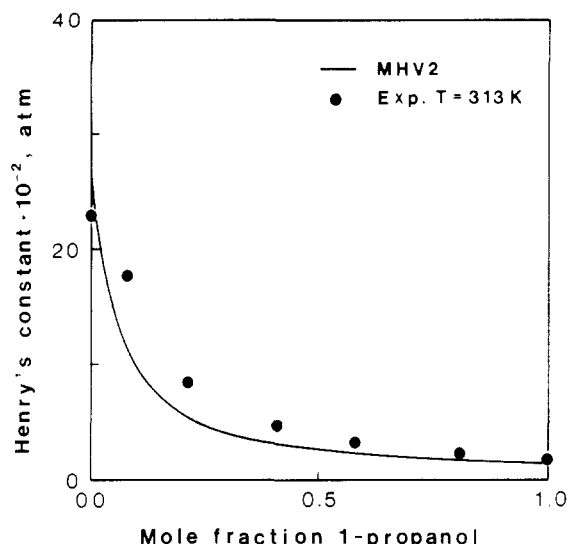


Figure 5. Experimental and predicted Henry's constants for CO_2 in 1-propanol-water solutions at 313 K. Experimental data: Tokunaga (1975).

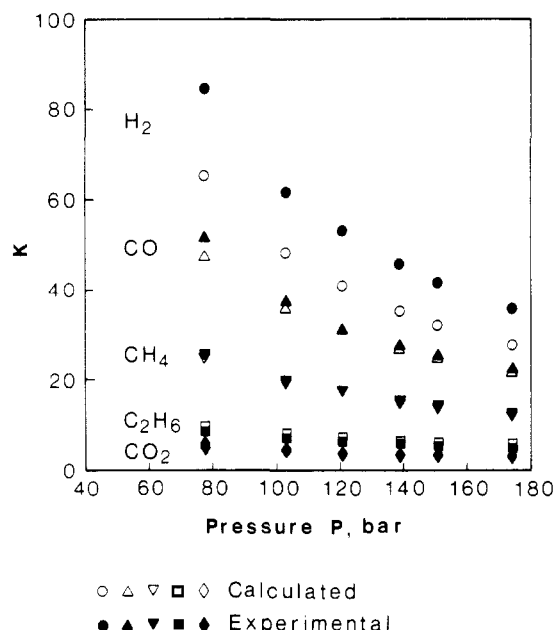


Figure 6. Experimental and predicted K values of the 10 component system: $\text{H}_2\text{-CO-CO}_2\text{-CH}_4\text{-C}_2\text{H}_6\text{-C}_3\text{H}_8\text{-H}_2\text{O-MeOH-EtOH-1-PrOH}$. The calculated K values (y/x) were obtained from PT flash calculations. Experimental data: Suzuki et al. (1990).

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Nomenclature

a = equation of state mixture parameter
 a_{ij} = interaction energy parameter of UNIFAC(K)
 b = equation of state mixture parameter
 C = number of components
 C_1, C_2, C_3 = pure-component temperature-dependence parameters
 f = fugacity
 g^E = excess Gibbs energy

P = pressure, bar
 q_1, q_2 = mixing rule constants (eq 7)
 Q = structural group or molecular surface area
 R_v = structural group or molecular volume
 R = gas constant
 T = temperature, K
 v = molar volume
 x = liquid mole fraction
 y = vapor mole fraction
 z = phase composition
 K = equilibrium ratio (y/x)

Superscripts

E = excess property
 l = liquid-phase property
 v = vapor-phase property

Subscripts

c = critical values
 ii = parameter for pure component i
 r = reduced property

Greek Symbols

α = equation of state mixture parameter, $\alpha = a/bRT$
 φ = fugacity coefficient
 ω = acentric factor

Appendix. Experimental Data and Objective Functions

Table X summarizes the experimental information used in the estimation of parameters. The table provides some indication of the expected quality of the representation for specific classes of systems. We have listed the number of different mixtures for which experimental data were used in the estimation of parameters, if there was more than one.

The gas-solvent group interaction parameters have been determined in the following order:

1. The parameters between the CH_2 group and the gases were estimated from gas plus alkane phase equilibrium data.
2. For each class of systems the remaining group-interaction parameters were determined. For instance, the parameters between H_2S and the CH_2 group were determined from data for the mixtures $\text{H}_2\text{S-hexane}$, $\text{H}_2\text{S-heptane}$, and $\text{H}_2\text{S-octane}$. Next, the interaction parameters between H_2S and the alcohol group (OH) were determined from data for mixtures with H_2S and alcohols using the $\text{CH}_2\text{-H}_2\text{S}$ parameters estimated earlier.

In the parameter estimations we have used three types of binary experimental data:

1. TPXY (A in Table X): X , the mole fraction in phase 1 (liquid); Y , the mole fraction in phase 2 measured as vapor-liquid equilibrium (VLE) data or liquid-liquid equilibrium (LLE) data, at temperature T and pressure P .
2. TPX (B in Table X).
3. TX, (C in Table X): x , gas solubilities at atmospheric pressure.

The data sets were restricted in the sense that data points near the critical point were excluded.

In case 1, for VLE data, the parameter estimations were performed by minimization of the explicit objective function

$$\text{SSQ} = \sum_{n=1}^{\text{dtps}} \sum_{i=1}^2 \left(X_i \frac{\varphi_i^l(T, P, X)}{\varphi_i^v(T, P, Y)} - Y_i \right)^2 \quad (\text{A1})$$

where T , P , and X and Y are the experimental temperature, pressure, and phase compositions and φ_i is the fugacity coefficient ($\varphi^l = f^l/(PX)$, $\varphi^v = f^v/(PY)$). This objective function can be obtained from the isofugacity criterion ($f^l = f^v$). When the mole fractions of the gas in the liquid are very small (X_{gas} below 0.01), the linear form of eq A1 is inadequate and was in our work replaced by the logarithmic objective function:

$$\text{SSQ} = \sum_{n=1}^{\text{dtps}} \sum_{i=1}^2 \left[\ln \left(\frac{X_i \varphi_i^l(T, P, X)}{Y_i \varphi_i^v(T, P, Y)} \right) \right]^2 \quad (\text{A2})$$

This objective function also takes into account the small amounts of solvent in the vapor phase that are neglected by the linear form. For LLE data, PT flash calculations are performed to obtain the compositions in the two liquid phases and as the objective function is used:

$$\text{SSQ} = \sum_{n=1}^{\text{dtps}} \sum_{i=1}^2 \left[\ln \frac{X_{i,1}^{\text{cal}} X_{i,2}^{\text{cal}}}{X_{i,1}^{\text{exp}} X_{i,2}^{\text{exp}}} \right]^2 \quad (\text{A3})$$

where indexes 1 and 2 indicate liquid phases.

For the systems listed as A in Table X (TPXY) the parameters are applicable in the temperature range 273–523 K.

In cases 2 and 3, we have performed PT flash calculations to obtain the liquid-phase composition, and we use the objective function

$$\text{SSQ} = \sum_{n=1}^{\text{dtps}} \sum_{i=1}^2 \left[\ln \frac{X_i^{\text{cal}}}{X_i^{\text{exp}}} \right]^2 \quad (\text{A4})$$

The data for cases 2 and 3 do not always permit determination of two independent interaction parameters per binary. When a strong correlation between a_{ij} and a_{ji} is observed, a single interaction coefficient $a_{ij} = a_{ji}$ was used instead.

In case 2 we recommend the same temperature range for the use of parameters as in case 1. As type 3 data contain very limited information, parameters based on such data should be used with caution.

Below we have listed the references to data used in the estimation of parameters:

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Combination of Microtomy and NMR Spectroscopy To Probe the Variation in Horizontal Layers of Bondlines

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A technique was developed that provides information on the chloroform-soluble components in the horizontal layers of bondlines. The bondlines examined were in samples that consisted of a mat of polyaramid fibers, in an epoxy prepreg matrix, bonded to a layer of rubber. The sectioning, with a microtome instrument, of the bondlines (ca. 250 μm in thickness) yielded slices 20 μm in thickness. Approximately 12 slices/bondline were obtained. The chloroform-soluble fraction of these slices was analyzed by proton NMR spectroscopy to determine the chemical species present. This information was combined with the location of the slices to give a horizontal analysis of the concentrations of the different chloroform-soluble components present across the thickness of the bondline. The results obtained indicated the degree of cure of the different components of the original bondline, the migration of these components, and the migration of the plasticizer dioctyl phthalate (DOP) from the rubber layer into the bondline.

Introduction

Analysis of the chemistry of adhesive bondlines has relied on several methods to determine the degree of cure and chemical composition of the cured adhesive. The degree of cure can be measured by the quantity of soluble, uncured material present after cure and by other methods such as differential scanning calorimetry and dynamic mechanical analysis. Chemical tests can quantify the amount of unreacted epoxy groups and other functionalities present after curing. Infrared and NMR spectroscopy, chromatography, and elemental analysis are also useful tools for bondline analysis. Reviews have appeared recently by Jahn and Goetzky (1988) and Hadad (1988) of the various analytical methods available. However, more detailed information concerning selective migration of different components of the adhesive, migration of plasticizers from the materials bonded together into the bondline, and the degree of cure in different horizontal layers of the bondline is not directly obtained by the above methods.

A recent manufacturing investigation required the detailed information described above. We developed a combined technique of microtomy and NMR spectroscopy in the analysis of laboratory bondline samples to give the detailed description of bondlines we required. Microtomy is routinely used in many applications that require the sectioning of solid materials into very thin slices (ca. 1-50 μm in thickness) for further analysis. Ultramicrotomy, producing slices of ca. 300-600 Å in thickness, coupled with transmission electron microscopy has been utilized to examine the polymer-metal interface in composite-coated metal samples by Dartyge and Ferrieres (1980) and polymer-metal bonds by Schaefer et al. (1986). Similar analytical techniques have been employed by Wolf (1978) to measure structure, shape, and flaws for quality control in

plastics manufacturing and by Stangl et al. (1976) to evaluate the aging of high-density polyethylene and ABS copolymer samples at different depths. Also rubber blendability has been analyzed with these techniques by Kruse (1976).

We utilized standard microtomy on bondline samples to give individual horizontal slices 20 μm in thickness. The thinness of these slices made possible the collection of a sufficient number of samples per bondline in order to be able to obtain a detailed cross-sectional picture of the chemical species present. NMR analysis of the chloroform-soluble fraction of the slices provided the concentrations of the different soluble components. The studies reported herein involve both the development of the technique and the results obtained.

Experimental Section

Preparations of Bondline Samples. The bondline samples were prepared at Advanced Technology, Thiokol. Diagrams representing the configurations of the samples are presented in Figure 1. Flat panels (15 cm $\times$ 15 cm $\times$ 1.3 cm) of Kevlar (a registered trademark of Dupont) fibers, containing an epoxy prepreg matrix (resin 1), were laminated, and one side was coated with ca. 0.25-mm layer of an additional amount of resin 1. A second different adhesive layer (resin 2) (held in a Dacron (a registered trademark of Dupont) mesh) was placed on top of the layer of resin 1. This two-resin system (resins 1 and 2) gives a better bond than just resin 1 by itself. The next layer of the samples consisted of V-45 NBR rubber (Kirkhill) (2.5 mm in thickness). The rubber layers employed were preconditioned to give samples with two different moisture contents. This was accomplished by preconditioning at 0 and 60% relative humidity for 24 h, respectively. Half of one side of each rubber layer (both "dry" and "wet"